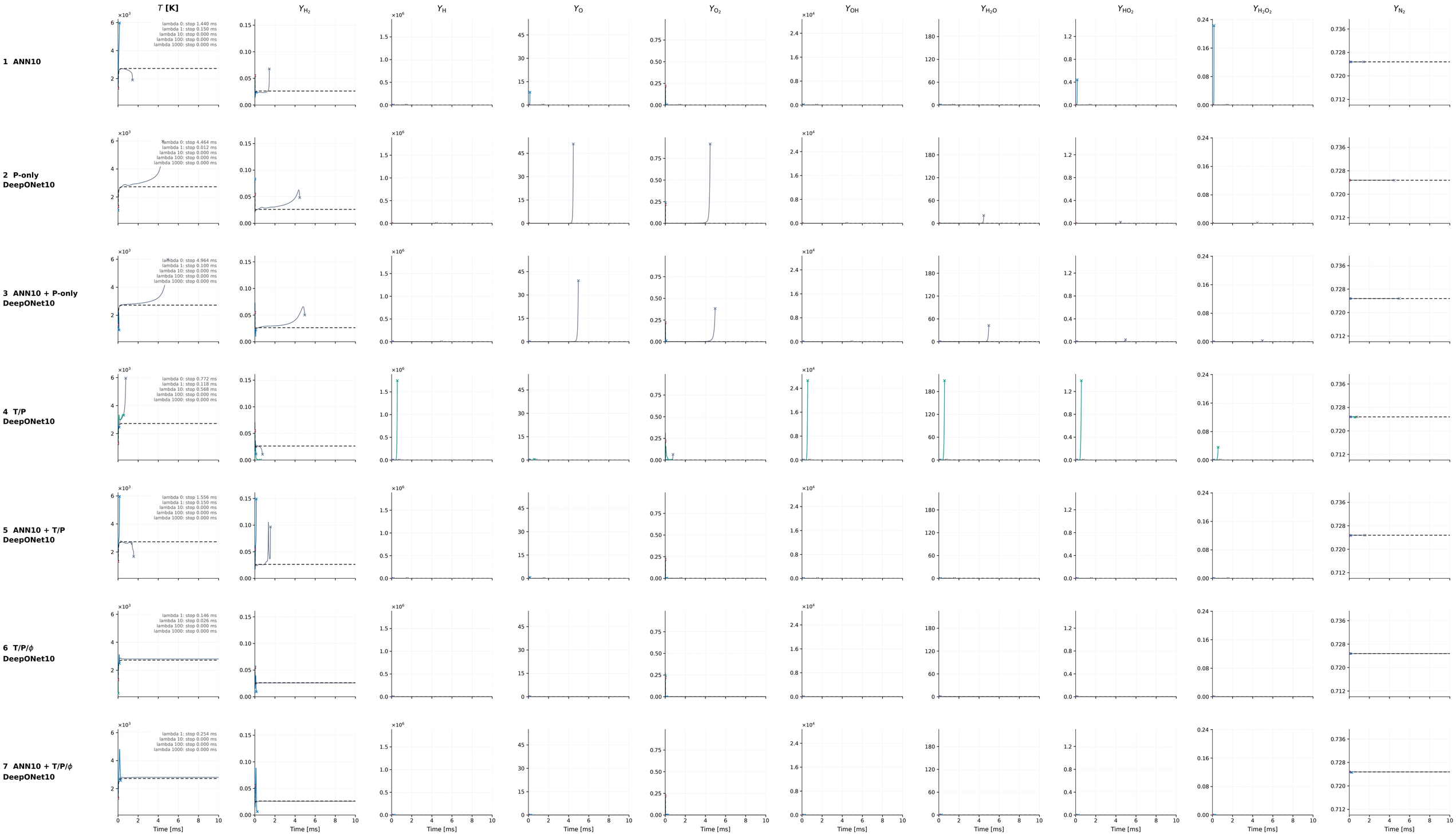
Element soft-constraint weight sweep | Test IC 04

T_0 = 1317.5 K P_0 = 1.244 atm ϕ_0 = 1.984 | Raw self-fed rollout, $\Delta t = 2\,\mu s$





Shared linear y scale within each variable column. Saved 2 us samples; early stops marked x.
Every member has10000 updates and the same seed-specific initialization/minibatch stream. Mixed teams use ANN10 + DeepONet10 at the same lambda. Lambda0/10 are frozen controls; inference is unchanged.

Lambda sweep | Test IC 04 | Cantera vs NN NRMSE

T0 = 1317.5 K P0 = 1.244 atm phi0 = 1.984 | Raw self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	lambda	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 1.440 ms
1 ANN10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.150 ms
1 ANN10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
1 ANN10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
1 ANN10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 4.464 ms
2 P-only DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.012 ms
2 P-only DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 4.964 ms
3 ANN10 + P-only DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.100 ms
3 ANN10 + P-only DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 0.772 ms
4 T/P DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.118 ms
4 T/P DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.568 ms
4 T/P DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 1.556 ms
5 ANN10 + T/P DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.150 ms
5 ANN10 + T/P DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	**0**	0.0309	0.0188	0.215	0.559	0.164	0.406	0.0425	0.406	0.363	6.13e-16	Complete
6 T/P/ϕ DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.146 ms
6 T/P/ϕ DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.026 ms
6 T/P/ϕ DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	**0**	0.0296	0.0176	0.223	0.634	0.106	0.427	0.047	0.294	0.389	1.53e-16	Complete
7 ANN10 + T/P/ϕ DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.254 ms
7 ANN10 + T/P/ϕ DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms

Exact user util.py defaults: norm="rms", reference=Cantera; equal time-sample weights, dimensionless ratio without x100.
N/A: undefined zero-reference RMS or empty common set. —: early stop, so no full-horizon NRMSE. Nonzero constant N2 is valid.
Every network has10000 updates. Lambda0/10 controls are reused; lambda1/100/1000 are fresh paired-seed models.
Every graph row includes all5 lambdas and the same Cantera reference.

Element soft-constraint weight sweep | Test IC 05

T_0 = 1049.0 K P_0 = 4.666 atm ϕ_0 = 0.674 | Raw self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference
lambda = 0 (10k)
lambda = 1 (10k)
lambda = 10 (10k)
lambda = 100 (10k)
lambda = 1000 (10k)

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

Shared linear y scale within each variable column. Saved 2 us samples; early stops marked x.
Every member has10000 updates and the same seed-specific initialization/minibatch stream. Mixed teams use ANN10 + DeepONet10 at the same lambda. Lambda0/10 are frozen controls; inference is unchanged.

Lambda sweep | Test IC 05 | Cantera vs NN NRMSE

T0 = 1049.0 K P0 = 4.666 atm phi0 = 0.674 | Raw self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

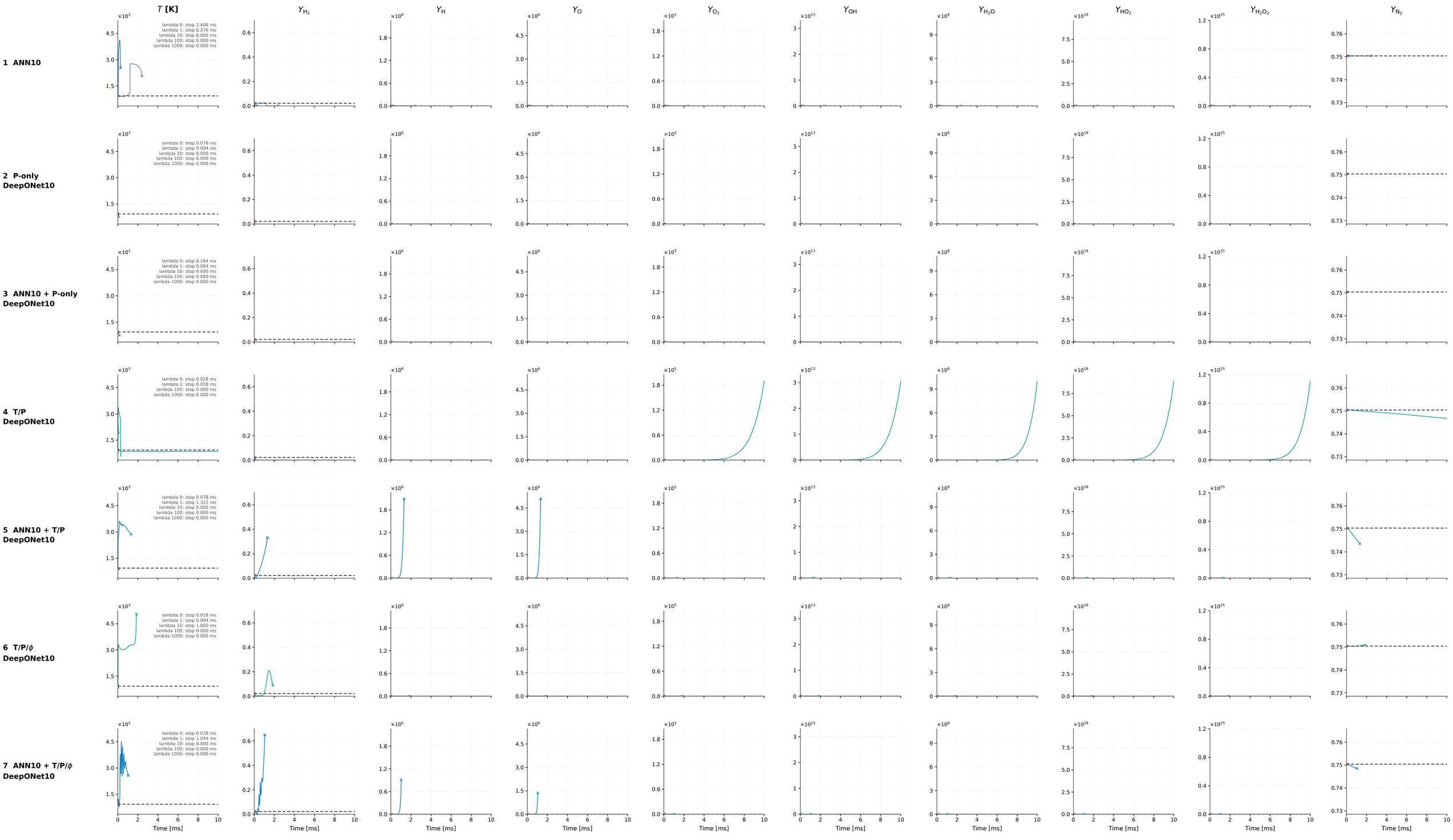
Model / same-state mean	lambda	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 1.660 ms
1 ANN10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.192 ms
1 ANN10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
1 ANN10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
1 ANN10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 0.186 ms
2 P-only DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.008 ms
2 P-only DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 0.544 ms
3 ANN10 + P-only DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.038 ms
3 ANN10 + P-only DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 0.420 ms
4 T/P DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.052 ms
4 T/P DeepONet10	**10**	0.669	0.986	1.48e+03	1.03e+14	4.87e+07	4.04e+15	8.00e+09	5.05e+22	5.88e+28	5.10e-04	Complete
4 T/P DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 0.950 ms
5 ANN10 + T/P DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.990 ms
5 ANN10 + T/P DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	**0**	1.71	0.835	29.8	1.22e+03	16.9	336	0.817	2.37	0.961	1.48e-16	Complete
6 T/P/ϕ DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.386 ms
6 T/P/ϕ DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 1.122 ms
6 T/P/ϕ DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	**0**	1.5	0.903	68.8	2.19e+03	16.8	203	0.602	2.85	0.951	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.866 ms
7 ANN10 + T/P/ϕ DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms

Exact user util.py defaults: norm="rms", reference=Cantera; equal time-sample weights, dimensionless ratio without x100.
N/A: undefined zero-reference RMS or empty common set. —: early stop, so no full-horizon NRMSE. Nonzero constant N2 is valid.
Every network has10000 updates. Lambda0/10 controls are reused; lambda1/100/1000 are fresh paired-seed models.
Every graph row includes all5 lambdas and the same Cantera reference.

Element soft-constraint weight sweep | Test IC 07

T_0 = 930.6 K P_0 = 3.722 atm ϕ_0 = 0.754 | Raw self-fed rollout, $\Delta t = 2\,\mu$s





Shared linear y scale within each variable column. Saved 2 us samples; early stops marked x.
Every member has10000 updates and the same seed-specific initialization/minibatch stream. Mixed teams use ANN10 + DeepONet10 at the same lambda. Lambda0/10 are frozen controls; inference is unchanged.

Lambda sweep | Test IC 07 | Cantera vs NN NRMSE

T0 = 930.6 K P0 = 3.722 atm phi0 = 0.754 | Raw self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	lambda	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 2.406 ms
1 ANN10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.270 ms
1 ANN10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
1 ANN10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
1 ANN10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 0.076 ms
2 P-only DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.004 ms
2 P-only DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 0.164 ms
3 ANN10 + P-only DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.004 ms
3 ANN10 + P-only DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 0.028 ms
4 T/P DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.058 ms
4 T/P DeepONet10	**10**	0.368	0.996	1.76e+09	1.62e+14	2.02e+05	1.57e+24	1.23e+15	4.06e+24	6.11e+30	2.67e-03	Complete
4 T/P DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 0.078 ms
5 ANN10 + T/P DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 1.322 ms
5 ANN10 + T/P DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 0.018 ms
6 T/P/ϕ DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.004 ms
6 T/P/ϕ DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 1.860 ms
6 T/P/ϕ DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 0.078 ms
7 ANN10 + T/P/ϕ DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 1.044 ms
7 ANN10 + T/P/ϕ DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms

Exact user util.py defaults: norm="rms", reference=Cantera; equal time-sample weights, dimensionless ratio without x100.
N/A: undefined zero-reference RMS or empty common set. —: early stop, so no full-horizon NRMSE. Nonzero constant N2 is valid.
Every network has10000 updates. Lambda0/10 controls are reused; lambda1/100/1000 are fresh paired-seed models.
Every graph row includes all5 lambdas and the same Cantera reference.

Element soft-constraint weight sweep | Test IC 18

T_0 = 1069.1 K P_0 = 2.170 atm ϕ_0 = 1.544 | Raw self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference
lambda = 0 (10k)
lambda = 1 (10k)
lambda = 10 (10k)
lambda = 100 (10k)
lambda = 1000 (10k)

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] | Y_{H_2} | Y_H | Y_O | Y_{O_2} | Y_{OH} | Y_{H_2O} | Y_{HO_2} | $Y_{H_2O_2}$ | Y_{N_2}

Time [ms]

Shared linear y scale within each variable column. Saved 2 us samples; early stops marked x.
Every member has10000 updates and the same seed-specific initialization/minibatch stream. Mixed teams use ANN10 + DeepONet10 at the same lambda. Lambda0/10 are frozen controls; inference is unchanged.

Lambda sweep | Test IC 18 | Cantera vs NN NRMSE

T0 = 1069.1 K P0 = 2.170 atm phi0 = 1.544 | Raw self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

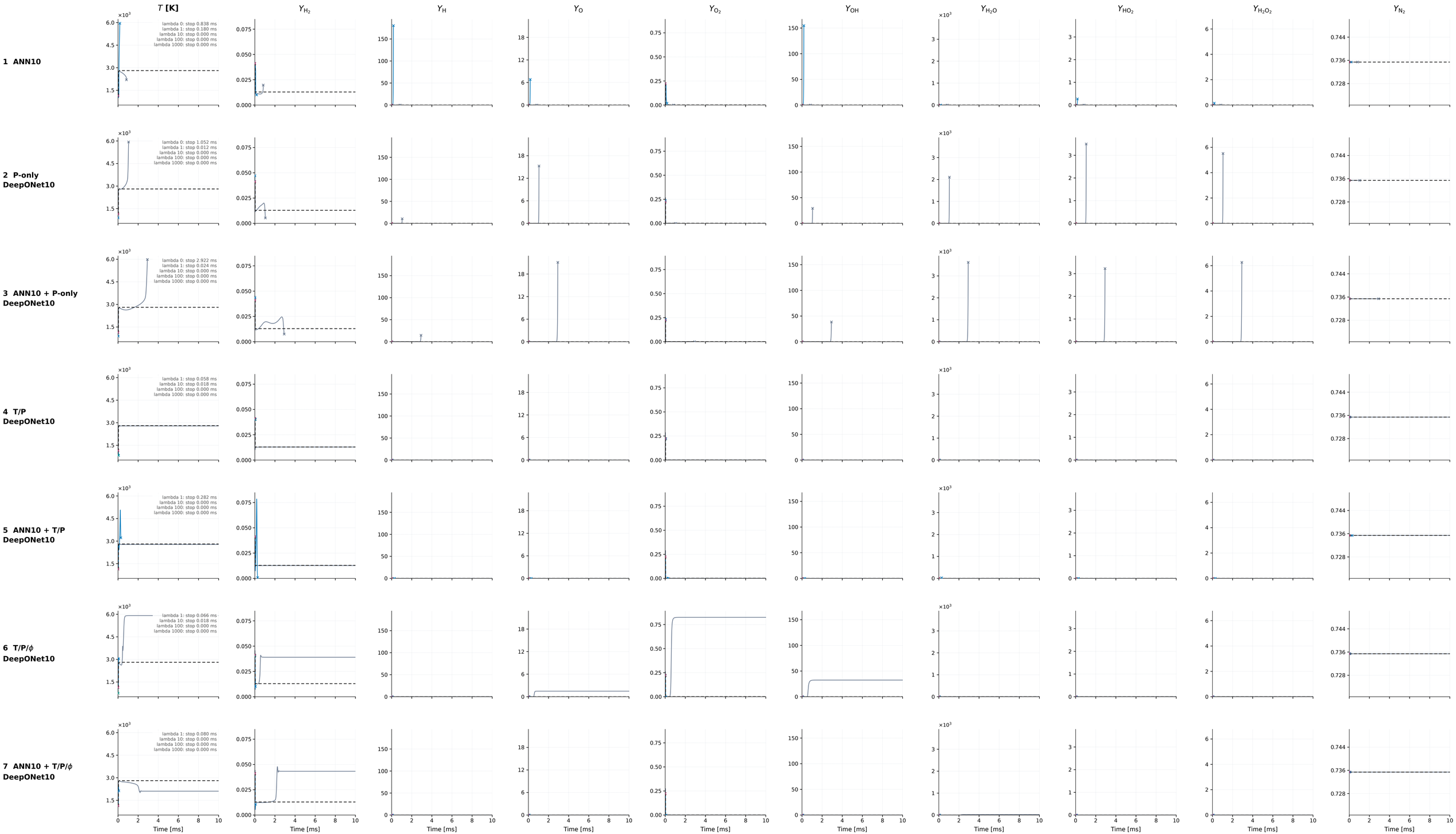
Model / same-state mean	lambda	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 1.382 ms
1 ANN10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.146 ms
1 ANN10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
1 ANN10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
1 ANN10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 1.124 ms
2 P-only DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.142 ms
2 P-only DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 4.796 ms
3 ANN10 + P-only DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.226 ms
3 ANN10 + P-only DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 0.108 ms
4 T/P DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.280 ms
4 T/P DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 1.000 ms
4 T/P DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 0.198 ms
5 ANN10 + T/P DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.186 ms
5 ANN10 + T/P DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 0.070 ms
6 T/P/ϕ DeepONet10	**1**	0.101	292	1.81e+23	1.09e+24	1.24e+10	3.21e+27	282	8.47e+22	3.71e+22	0.0429	Complete
6 T/P/ϕ DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.626 ms
6 T/P/ϕ DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 0.144 ms
7 ANN10 + T/P/ϕ DeepONet10	**1**	0.286	56.1	1.82e+22	8.33e+22	2.41e+11	3.26e+26	0.991	1.43e+21	5.75e+21	0.0105	Complete
7 ANN10 + T/P/ϕ DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms

Exact user util.py defaults: norm="rms", reference=Cantera; equal time-sample weights, dimensionless ratio without x100.
N/A: undefined zero-reference RMS or empty common set. —: early stop, so no full-horizon NRMSE. Nonzero constant N2 is valid.
Every network has10000 updates. Lambda0/10 controls are reused; lambda1/100/1000 are fresh paired-seed models.
Every graph row includes all5 lambdas and the same Cantera reference.

Element soft-constraint weight sweep | Test IC 32

T_0 = 1140.7 K P_0 = 3.354 atm ϕ_0 = 1.463 | Raw self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference
lambda = 0 (10k)
lambda = 1 (10k)
lambda = 10 (10k)
lambda = 100 (10k)
lambda = 1000 (10k)



Shared linear y scale within each variable column. Saved 2 us samples; early stops marked x.
Every member has10000 updates and the same seed-specific initialization/minibatch stream. Mixed teams use ANN10 + DeepONet10 at the same lambda. Lambda0/10 are frozen controls; inference is unchanged.

Lambda sweep | Test IC 32 | Cantera vs NN NRMSE

T0 = 1140.7 K P0 = 3.354 atm phi0 = 1.463 | Raw self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	lambda	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 0.838 ms
1 ANN10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.180 ms
1 ANN10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
1 ANN10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
1 ANN10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 1.052 ms
2 P-only DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.012 ms
2 P-only DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 2.922 ms
3 ANN10 + P-only DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.024 ms
3 ANN10 + P-only DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	**0**	0.0202	0.0859	0.158	0.382	0.588	0.0956	0.0328	1.22	0.997	1.06e-15	Complete
4 T/P DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.058 ms
4 T/P DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.018 ms
4 T/P DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	**0**	0.0163	0.0579	0.204	0.378	0.395	0.172	0.0257	1.12	1.1	1.06e-15	Complete
5 ANN10 + T/P DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.282 ms
5 ANN10 + T/P DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	**0**	1.07	1.94	2.39	2.27e+03	58	3.86e+03	0.953	23.8	421	1.06e-15	Complete
6 T/P/ϕ DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.066 ms
6 T/P/ϕ DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.018 ms
6 T/P/ϕ DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	**0**	0.224	2.05	102	0.727	0.326	0.829	75.3	1.85	30.6	1.06e-15	Complete
7 ANN10 + T/P/ϕ DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.080 ms
7 ANN10 + T/P/ϕ DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms

Exact user util.py defaults: norm="rms", reference=Cantera; equal time-sample weights, dimensionless ratio without x100.
N/A: undefined zero-reference RMS or empty common set. —: early stop, so no full-horizon NRMSE. Nonzero constant N2 is valid.
Every network has10000 updates. Lambda0/10 controls are reused; lambda1/100/1000 are fresh paired-seed models.
Every graph row includes all5 lambdas and the same Cantera reference.

Element soft-constraint weight sweep | Test IC 39

T_0 = 1250.1 K P_0 = 2.297 atm ϕ_0 = 0.646 | Raw self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference
lambda = 0 (10k)
lambda = 1 (10k)
lambda = 10 (10k)
lambda = 100 (10k)
lambda = 1000 (10k)

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

Shared linear y scale within each variable column. Saved 2 us samples; early stops marked x.
Every member has10000 updates and the same seed-specific initialization/minibatch stream. Mixed teams use ANN10 + DeepONet10 at the same lambda. Lambda0/10 are frozen controls; inference is unchanged.

Lambda sweep | Test IC 39 | Cantera vs NN NRMSE

T0 = 1250.1 K P0 = 2.297 atm phi0 = 0.646 | Raw self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	lambda	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 2.082 ms
1 ANN10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.072 ms
1 ANN10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
1 ANN10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
1 ANN10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 0.704 ms
2 P-only DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.014 ms
2 P-only DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 1.534 ms
3 ANN10 + P-only DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.038 ms
3 ANN10 + P-only DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 0.228 ms
4 T/P DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.092 ms
4 T/P DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.018 ms
4 T/P DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	**0**	0.225	0.427	126	12	214	167	67.7	97.8	29.2	5.90e-16	Complete
5 ANN10 + T/P DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.416 ms
5 ANN10 + T/P DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 0.334 ms
6 T/P/ϕ DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.080 ms
6 T/P/ϕ DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.018 ms
6 T/P/ϕ DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 0.452 ms
7 ANN10 + T/P/ϕ DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 7.218 ms
7 ANN10 + T/P/ϕ DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms

Exact user util.py defaults: norm="rms", reference=Cantera; equal time-sample weights, dimensionless ratio without x100.
N/A: undefined zero-reference RMS or empty common set. —: early stop, so no full-horizon NRMSE. Nonzero constant N2 is valid.
Every network has10000 updates. Lambda0/10 controls are reused; lambda1/100/1000 are fresh paired-seed models.
Every graph row includes all5 lambdas and the same Cantera reference.

Element soft-constraint weight sweep | Test IC 44

T_0 = 1031.1 K P_0 = 2.272 atm ϕ_0 = 0.770 | Raw self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference
lambda = 0 (10k)
lambda = 1 (10k)
lambda = 10 (10k)
lambda = 100 (10k)
lambda = 1000 (10k)

1 ANN10

2 P-only DeepONet10

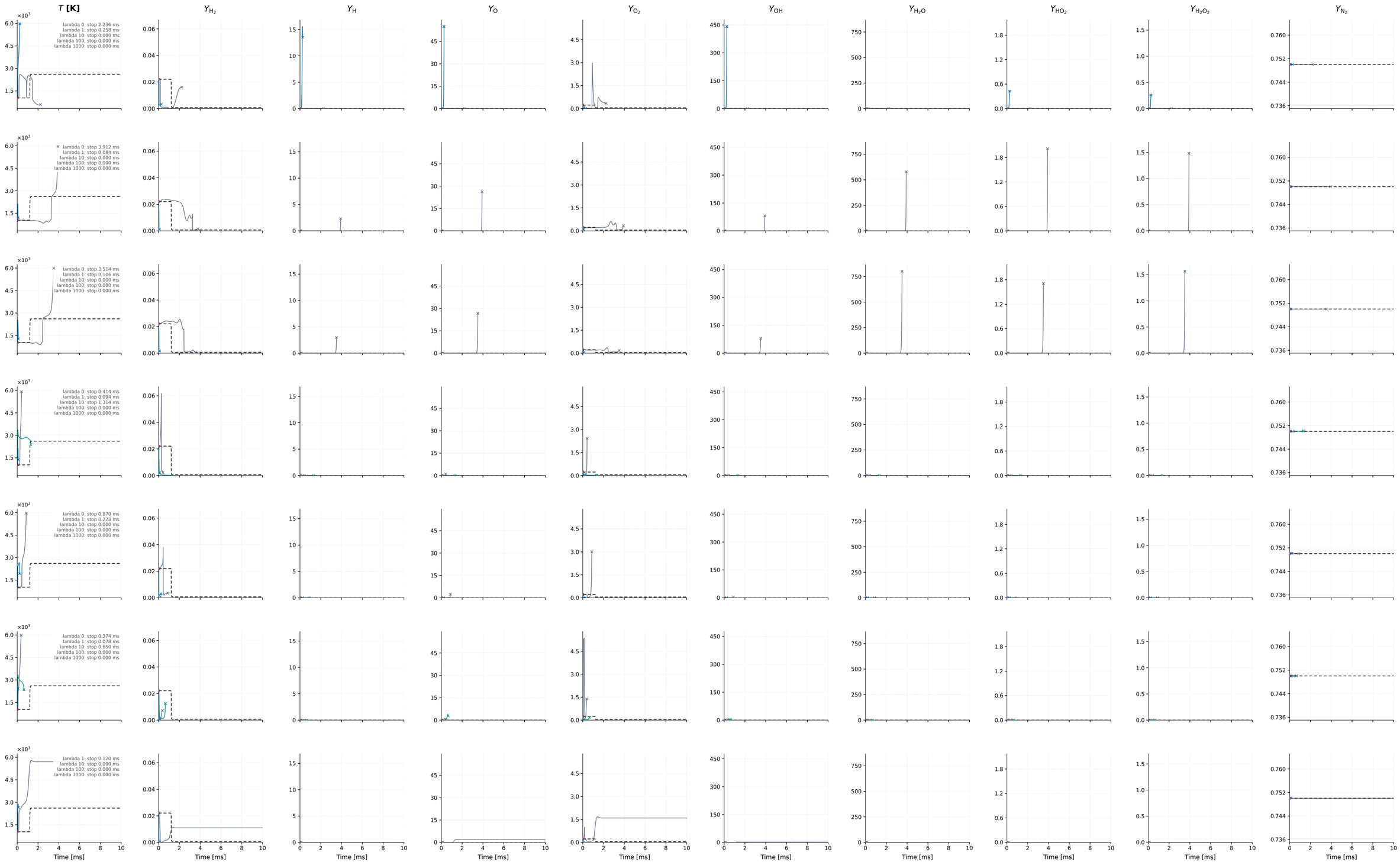
3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10



Shared linear y scale within each variable column. Saved 2 us samples; early stops marked x.
Every member has10000 updates and the same seed-specific initialization/minibatch stream. Mixed teams use ANN10 + DeepONet10 at the same lambda. Lambda0/10 are frozen controls; inference is unchanged.

Lambda sweep | Test IC 44 | Cantera vs NN NRMSE

T0 = 1031.1 K P0 = 2.272 atm phi0 = 0.770 | Raw self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	lambda	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 2.236 ms
1 ANN10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.258 ms
1 ANN10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
1 ANN10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
1 ANN10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 3.912 ms
2 P-only DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.084 ms
2 P-only DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 3.514 ms
3 ANN10 + P-only DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.106 ms
3 ANN10 + P-only DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 0.414 ms
4 T/P DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.094 ms
4 T/P DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 1.314 ms
4 T/P DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 0.870 ms
5 ANN10 + T/P DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.228 ms
5 ANN10 + T/P DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 0.374 ms
6 T/P/ϕ DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.078 ms
6 T/P/ϕ DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.650 ms
6 T/P/ϕ DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	**0**	1.21	1.51	31.7	859	15.7	160	0.333	6.13	5.11	2.22e-15	Complete
7 ANN10 + T/P/ϕ DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.120 ms
7 ANN10 + T/P/ϕ DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms

Exact user util.py defaults: norm="rms", reference=Cantera; equal time-sample weights, dimensionless ratio without x100.
N/A: undefined zero-reference RMS or empty common set. —: early stop, so no full-horizon NRMSE. Nonzero constant N2 is valid.
Every network has10000 updates. Lambda0/10 controls are reused; lambda1/100/1000 are fresh paired-seed models.
Every graph row includes all5 lambdas and the same Cantera reference.

Element soft-constraint weight sweep | Test IC 45

T_0 = 1181.9 K P_0 = 4.842 atm ϕ_0 = 1.099 | Raw self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference
lambda = 0 (10k)
lambda = 1 (10k)
lambda = 10 (10k)
lambda = 100 (10k)
lambda = 1000 (10k)

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

Shared linear y scale within each variable column. Saved 2 us samples; early stops marked x.
Every member has10000 updates and the same seed-specific initialization/minibatch stream. Mixed teams use ANN10 + DeepONet10 at the same lambda. Lambda0/10 are frozen controls; inference is unchanged.

Lambda sweep | Test IC 45 | Cantera vs NN NRMSE

T0 = 1181.9 K P0 = 4.842 atm phi0 = 1.099 | Raw self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	lambda	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**0**	0.154	1.63	0.954	0.964	0.621	0.93	0.638	0.705	0.788	1.49e-16	Complete
1 ANN10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.158 ms
1 ANN10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
1 ANN10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
1 ANN10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 0.352 ms
2 P-only DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.036 ms
2 P-only DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 0.850 ms
3 ANN10 + P-only DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.040 ms
3 ANN10 + P-only DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	**0**	0.0217	0.262	0.261	0.199	0.511	0.0576	0.0293	0.84	0.605	1.49e-16	Complete
4 T/P DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.100 ms
4 T/P DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.010 ms
4 T/P DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	**0**	0.0214	0.315	0.312	0.16	0.505	0.116	0.0315	0.885	0.653	1.49e-16	Complete
5 ANN10 + T/P DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 1.202 ms
5 ANN10 + T/P DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	**0**	0.897	2.9	2.57	818	1.26e+03	788	5.08	31.5	155	1.49e-16	Complete
6 T/P/ϕ DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.454 ms
6 T/P/ϕ DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.012 ms
6 T/P/ϕ DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	**0**	0.805	3.6	3.74	1.03e+03	80.1	212	0.391	12.5	67.8	1.49e-16	Complete
7 ANN10 + T/P/ϕ DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.760 ms
7 ANN10 + T/P/ϕ DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms

Exact user util.py defaults: norm="rms", reference=Cantera; equal time-sample weights, dimensionless ratio without x100.
N/A: undefined zero-reference RMS or empty common set. —: early stop, so no full-horizon NRMSE. Nonzero constant N2 is valid.
Every network has10000 updates. Lambda0/10 controls are reused; lambda1/100/1000 are fresh paired-seed models.
Every graph row includes all5 lambdas and the same Cantera reference.

Element soft-constraint weight sweep | Test IC 53

T_0 = 1257.8 K P_0 = 2.103 atm ϕ_0 = 1.799 | Raw self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference
lambda = 0 (10k)
lambda = 1 (10k)
lambda = 10 (10k)
lambda = 100 (10k)
lambda = 1000 (10k)

1 ANN10
lambda 0: stop 1.320 ms
lambda 1: stop 0.176 ms
lambda 10: stop 0.000 ms
lambda 100: stop 0.000 ms
lambda 1000: stop 0.000 ms

2 P-only DeepONet10
lambda 0: stop 2.148 ms
lambda 1: stop 0.012 ms
lambda 10: stop 0.000 ms
lambda 100: stop 0.000 ms
lambda 1000: stop 0.000 ms

3 ANN10 + P-only DeepONet10
lambda 1: stop 0.034 ms
lambda 10: stop 0.000 ms
lambda 100: stop 0.000 ms
lambda 1000: stop 0.000 ms

4 T/P DeepONet10
lambda 1: stop 0.122 ms
lambda 10: stop 0.020 ms
lambda 100: stop 0.000 ms
lambda 1000: stop 0.000 ms

5 ANN10 + T/P DeepONet10
lambda 1: stop 0.178 ms
lambda 10: stop 0.000 ms
lambda 100: stop 0.000 ms
lambda 1000: stop 0.000 ms

6 T/P/ϕ DeepONet10
lambda 0: stop 0.294 ms
lambda 10: stop 0.020 ms
lambda 100: stop 0.000 ms
lambda 1000: stop 0.000 ms

7 ANN10 + T/P/ϕ DeepONet10
lambda 10: stop 0.000 ms
lambda 100: stop 0.000 ms
lambda 1000: stop 0.000 ms

Columns: T [K], Y_{H_2}, Y_H, Y_O, Y_{O_2}, Y_{OH}, Y_{H_2O}, Y_{HO_2}, $Y_{H_2O_2}$, Y_{N_2}

Time [ms]

Shared linear y scale within each variable column. Saved 2 us samples; early stops marked x.
Every member has10000 updates and the same seed-specific initialization/minibatch stream. Mixed teams use ANN10 + DeepONet10 at the same lambda. Lambda0/10 are frozen controls; inference is unchanged.

Lambda sweep | Test IC 53 | Cantera vs NN NRMSE

T0 = 1257.8 K P0 = 2.103 atm phi0 = 1.799 | Raw self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	lambda	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 1.320 ms
1 ANN10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.176 ms
1 ANN10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
1 ANN10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
1 ANN10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 2.148 ms
2 P-only DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.012 ms
2 P-only DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	**0**	0.0124	0.292	0.396	0.464	0.0498	0.204	0.136	0.121	0.506	7.62e-16	Complete
3 ANN10 + P-only DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.034 ms
3 ANN10 + P-only DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	**0**	0.0377	0.0241	0.286	0.477	0.312	0.332	0.0163	0.592	0.456	7.62e-16	Complete
4 T/P DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.122 ms
4 T/P DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.020 ms
4 T/P DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	**0**	0.0356	0.0124	0.425	0.598	0.201	0.426	0.106	0.415	0.373	7.62e-16	Complete
5 ANN10 + T/P DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.178 ms
5 ANN10 + T/P DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 0.294 ms
6 T/P/ϕ DeepONet10	**1**	0.147	360	9.33e+23	1.08e+25	1.37e+13	3.34e+28	1.32	1.87e+24	2.05e+24	0.0195	Complete
6 T/P/ϕ DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.020 ms
6 T/P/ϕ DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	**0**	0.05	0.249	0.583	0.893	0.258	0.86	0.241	0.547	1.1	7.62e-16	Complete
7 ANN10 + T/P/ϕ DeepONet10	**1**	0.465	1.17e+04	3.18e+22	8.21e+23	3.06e+14	8.90e+26	0.925	8.46e+21	4.97e+22	4.06e-03	Complete
7 ANN10 + T/P/ϕ DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms

Exact user util.py defaults: norm="rms", reference=Cantera; equal time-sample weights, dimensionless ratio without x100.
N/A: undefined zero-reference RMS or empty common set. —: early stop, so no full-horizon NRMSE. Nonzero constant N2 is valid.
Every network has10000 updates. Lambda0/10 controls are reused; lambda1/100/1000 are fresh paired-seed models.
Every graph row includes all5 lambdas and the same Cantera reference.

Element soft-constraint weight sweep | Test IC 57

T_0 = 1412.0 K P_0 = 4.083 atm ϕ_0 = 0.677 | Raw self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference
lambda = 0 (10k)
lambda = 1 (10k)
lambda = 10 (10k)
lambda = 100 (10k)
lambda = 1000 (10k)

1 ANN10
lambda 0: stop 2.010 ms
lambda 1: stop 0.092 ms
lambda 10: stop 0.000 ms
lambda 100: stop 0.000 ms
lambda 1000: stop 0.000 ms

2 P-only DeepONet10
lambda 0: stop 0.524 ms
lambda 1: stop 0.016 ms
lambda 10: stop 0.000 ms
lambda 100: stop 0.000 ms
lambda 1000: stop 0.000 ms

3 ANN10 + P-only DeepONet10
lambda 0: stop 0.954 ms
lambda 1: stop 0.024 ms
lambda 10: stop 0.000 ms
lambda 100: stop 0.000 ms
lambda 1000: stop 0.000 ms

4 T/P DeepONet10
lambda 0: stop 0.240 ms
lambda 1: stop 1.832 ms
lambda 10: stop 0.186 ms
lambda 100: stop 0.000 ms
lambda 1000: stop 0.000 ms

5 ANN10 + T/P DeepONet10
lambda 1: stop 1.100 ms
lambda 10: stop 0.000 ms
lambda 100: stop 0.000 ms
lambda 1000: stop 0.000 ms

6 T/P/ϕ DeepONet10
lambda 0: stop 0.452 ms
lambda 1: stop 0.080 ms
lambda 10: stop 0.352 ms
lambda 100: stop 0.000 ms
lambda 1000: stop 0.000 ms

7 ANN10 + T/P/ϕ DeepONet10
lambda 1: stop 0.774 ms
lambda 10: stop 0.000 ms
lambda 100: stop 0.000 ms
lambda 1000: stop 0.000 ms

T [K] | Y_{H_2} | Y_H | Y_O | Y_{O_2} | Y_{OH} | Y_{H_2O} | Y_{HO_2} | $Y_{H_2O_2}$ | Y_{N_2}

Time [ms]

Shared linear y scale within each variable column. Saved 2 us samples; early stops marked x.
Every member has10000 updates and the same seed-specific initialization/minibatch stream. Mixed teams use ANN10 + DeepONet10 at the same lambda. Lambda0/10 are frozen controls; inference is unchanged.

Lambda sweep | Test IC 57 | Cantera vs NN NRMSE

T0 = 1412.0 K P0 = 4.083 atm phi0 = 0.677 | Raw self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	lambda	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 2.010 ms
1 ANN10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.092 ms
1 ANN10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
1 ANN10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
1 ANN10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 0.524 ms
2 P-only DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.016 ms
2 P-only DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 0.954 ms
3 ANN10 + P-only DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.024 ms
3 ANN10 + P-only DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 0.240 ms
4 T/P DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 1.832 ms
4 T/P DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.186 ms
4 T/P DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	**0**	0.332	0.813	56.4	0.932	63.3	136	59.5	57.4	32.6	1.48e-16	Complete
5 ANN10 + T/P DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 1.100 ms
5 ANN10 + T/P DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 0.452 ms
6 T/P/ϕ DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.080 ms
6 T/P/ϕ DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.352 ms
6 T/P/ϕ DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	**0**	0.977	9.31	18.2	665	32.5	95.2	0.146	6.37	26.8	1.48e-16	Complete
7 ANN10 + T/P/ϕ DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.774 ms
7 ANN10 + T/P/ϕ DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms

Exact user util.py defaults: norm="rms", reference=Cantera; equal time-sample weights, dimensionless ratio without x100.
N/A: undefined zero-reference RMS or empty common set. —: early stop, so no full-horizon NRMSE. Nonzero constant N2 is valid.
Every network has10000 updates. Lambda0/10 controls are reused; lambda1/100/1000 are fresh paired-seed models.
Every graph row includes all5 lambdas and the same Cantera reference.

Element soft-constraint weight sweep | Test IC 64

T_0 = 1045.3 K P_0 = 4.301 atm ϕ_0 = 0.595 | Raw self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference
lambda = 0 (10k)
lambda = 1 (10k)
lambda = 10 (10k)
lambda = 100 (10k)
lambda = 1000 (10k)

1 ANN10
lambda 0: stop 1.036 ms
lambda 1: stop 0.180 ms
lambda 10: stop 0.000 ms
lambda 100: stop 0.000 ms
lambda 1000: stop 0.000 ms

2 P-only DeepONet10
lambda 0: stop 0.148 ms
lambda 1: stop 0.006 ms
lambda 10: stop 0.000 ms
lambda 100: stop 0.000 ms
lambda 1000: stop 0.000 ms

3 ANN10 + P-only DeepONet10
lambda 0: stop 0.424 ms
lambda 1: stop 0.038 ms
lambda 10: stop 0.000 ms
lambda 100: stop 0.000 ms
lambda 1000: stop 0.000 ms

4 T/P DeepONet10
lambda 0: stop 0.432 ms
lambda 1: stop 0.054 ms
lambda 10: stop 0.210 ms
lambda 100: stop 0.000 ms
lambda 1000: stop 0.000 ms

5 ANN10 + T/P DeepONet10
lambda 0: stop 0.952 ms
lambda 1: stop 1.056 ms
lambda 10: stop 0.000 ms
lambda 100: stop 0.000 ms
lambda 1000: stop 0.000 ms

6 T/P/ϕ DeepONet10
lambda 0: stop 0.422 ms
lambda 1: stop 0.166 ms
lambda 10: stop 0.358 ms
lambda 100: stop 0.000 ms
lambda 1000: stop 0.000 ms

7 ANN10 + T/P/ϕ DeepONet10
lambda 1: stop 0.838 ms
lambda 10: stop 0.000 ms
lambda 100: stop 0.000 ms
lambda 1000: stop 0.000 ms

Columns: T [K], Y_{H_2}, Y_H, Y_O, Y_{O_2}, Y_{OH}, Y_{H_2O}, Y_{HO_2}, $Y_{H_2O_2}$, Y_{N_2}

Time [ms]

Shared linear y scale within each variable column. Saved 2 us samples; early stops marked x.
Every member has10000 updates and the same seed-specific initialization/minibatch stream. Mixed teams use ANN10 + DeepONet10 at the same lambda. Lambda0/10 are frozen controls; inference is unchanged.

Lambda sweep | Test IC 64 | Cantera vs NN NRMSE

T0 = 1045.3 K P0 = 4.301 atm phi0 = 0.595 | Raw self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	lambda	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 1.036 ms
1 ANN10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.180 ms
1 ANN10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
1 ANN10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
1 ANN10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 0.148 ms
2 P-only DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.006 ms
2 P-only DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 0.424 ms
3 ANN10 + P-only DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.038 ms
3 ANN10 + P-only DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 0.432 ms
4 T/P DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.054 ms
4 T/P DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.210 ms
4 T/P DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 0.952 ms
5 ANN10 + T/P DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 1.056 ms
5 ANN10 + T/P DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 0.422 ms
6 T/P/ϕ DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.166 ms
6 T/P/ϕ DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.358 ms
6 T/P/ϕ DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	**0**	1.72	0.836	120	3.17e+03	16.6	271	0.621	3.01	0.949	1.03e-15	Complete
7 ANN10 + T/P/ϕ DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.838 ms
7 ANN10 + T/P/ϕ DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms

Exact user util.py defaults: norm="rms", reference=Cantera; equal time-sample weights, dimensionless ratio without x100.
N/A: undefined zero-reference RMS or empty common set. —: early stop, so no full-horizon NRMSE. Nonzero constant N2 is valid.
Every network has10000 updates. Lambda0/10 controls are reused; lambda1/100/1000 are fresh paired-seed models.
Every graph row includes all5 lambdas and the same Cantera reference.

Element soft-constraint weight sweep | Test IC 68

T_0 = 1086.3 K P_0 = 3.702 atm ϕ_0 = 1.168 | Raw self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference
lambda = 0 (10k)
lambda = 1 (10k)
lambda = 10 (10k)
lambda = 100 (10k)
lambda = 1000 (10k)

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] | Y_{H_2} | Y_H | Y_O | Y_{O_2} | Y_{OH} | Y_{H_2O} | Y_{HO_2} | $Y_{H_2O_2}$ | Y_{N_2}

Time [ms]

Shared linear y scale within each variable column. Saved 2 us samples; early stops marked x.
Every member has10000 updates and the same seed-specific initialization/minibatch stream. Mixed teams use ANN10 + DeepONet10 at the same lambda. Lambda0/10 are frozen controls; inference is unchanged.

Lambda sweep | Test IC 68 | Cantera vs NN NRMSE

T0 = 1086.3 K P0 = 3.702 atm phi0 = 1.168 | Raw self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	lambda	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 1.008 ms
1 ANN10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.204 ms
1 ANN10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
1 ANN10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
1 ANN10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 2.764 ms
2 P-only DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.040 ms
2 P-only DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	**0**	0.295	1.67	0.514	0.978	5.87	0.866	0.624	2.66	163	4.49e-16	Complete
3 ANN10 + P-only DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.108 ms
3 ANN10 + P-only DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 0.422 ms
4 T/P DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.112 ms
4 T/P DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.634 ms
4 T/P DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 0.880 ms
5 ANN10 + T/P DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.574 ms
5 ANN10 + T/P DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	**0**	1.01	1.81	2.29	1.08e+03	26.3	575	0.851	7.61	18.2	4.49e-16	Complete
6 T/P/ϕ DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.096 ms
6 T/P/ϕ DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 1.404 ms
6 T/P/ϕ DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 0.872 ms
7 ANN10 + T/P/ϕ DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 1.304 ms
7 ANN10 + T/P/ϕ DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms

Exact user util.py defaults: norm="rms", reference=Cantera; equal time-sample weights, dimensionless ratio without x100.
N/A: undefined zero-reference RMS or empty common set. —: early stop, so no full-horizon NRMSE. Nonzero constant N2 is valid.
Every network has10000 updates. Lambda0/10 controls are reused; lambda1/100/1000 are fresh paired-seed models.
Every graph row includes all5 lambdas and the same Cantera reference.

Element soft-constraint weight sweep | Test IC 69

T_0 = 935.0 K P_0 = 2.045 atm ϕ_0 = 0.523 | Raw self-fed rollout, $\Delta t = 2\,\mu$s

Cantera reference
lambda = 0 (10k)
lambda = 1 (10k)
lambda = 10 (10k)
lambda = 100 (10k)
lambda = 1000 (10k)

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] | Y_{H_2} | Y_H | Y_O | Y_{O_2} | Y_{OH} | Y_{H_2O} | Y_{HO_2} | $Y_{H_2O_2}$ | Y_{N_2}

Time [ms]

Shared linear y scale within each variable column. Saved 2 us samples; early stops marked x.
Every member has10000 updates and the same seed-specific initialization/minibatch stream. Mixed teams use ANN10 + DeepONet10 at the same lambda. Lambda0/10 are frozen controls; inference is unchanged.

Lambda sweep | Test IC 69 | Cantera vs NN NRMSE

T0 = 935.0 K P0 = 2.045 atm phi0 = 0.523 | Raw self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

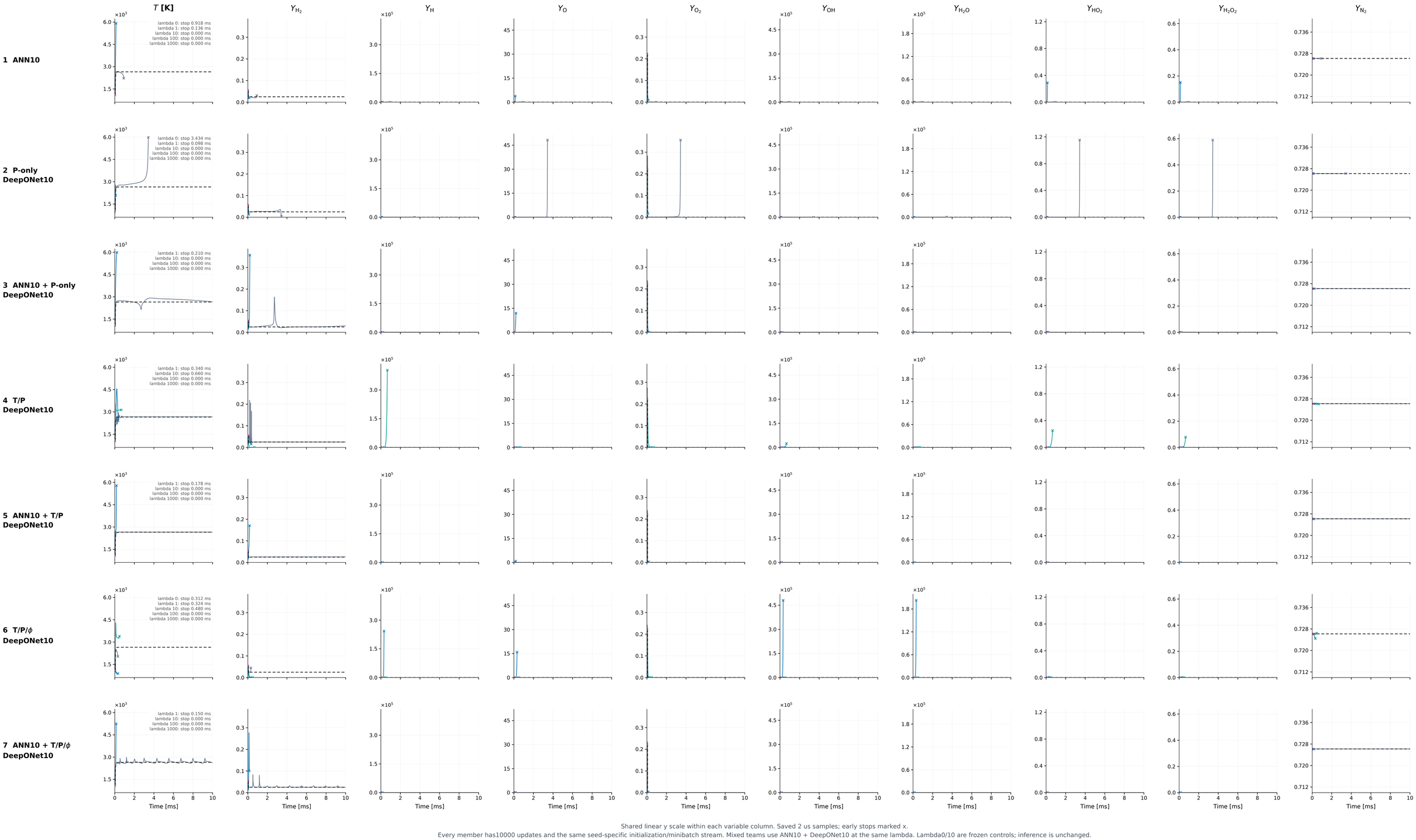
Model / same-state mean	lambda	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 0.128 ms
1 ANN10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.316 ms
1 ANN10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
1 ANN10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
1 ANN10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 0.208 ms
2 P-only DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.008 ms
2 P-only DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 0.280 ms
3 ANN10 + P-only DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.080 ms
3 ANN10 + P-only DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 0.046 ms
4 T/P DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.218 ms
4 T/P DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.010 ms
4 T/P DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 0.188 ms
5 ANN10 + T/P DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.074 ms
5 ANN10 + T/P DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 0.320 ms
6 T/P/ϕ DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.476 ms
6 T/P/ϕ DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 1.938 ms
6 T/P/ϕ DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	**0**	5.19	0.582	9.65e+08	1.69e+11	12.3	4.23e+11	2.17e+06	829	349	1.03e-15	Complete
7 ANN10 + T/P/ϕ DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.118 ms
7 ANN10 + T/P/ϕ DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms

Exact user util.py defaults: norm="rms", reference=Cantera; equal time-sample weights, dimensionless ratio without x100.
N/A: undefined zero-reference RMS or empty common set. —: early stop, so no full-horizon NRMSE. Nonzero constant N2 is valid.
Every network has10000 updates. Lambda0/10 controls are reused; lambda1/100/1000 are fresh paired-seed models.
Every graph row includes all5 lambdas and the same Cantera reference.

Element soft-constraint weight sweep | Test IC 73

T_0 = 1106.0 K P_0 = 1.911 atm ϕ_0 = 1.915 | Raw self-fed rollout, $\Delta t = 2\,\mu s$





Shared linear y scale within each variable column. Saved 2 us samples; early stops marked x.
Every member has10000 updates and the same seed-specific initialization/minibatch stream. Mixed teams use ANN10 + DeepONet10 at the same lambda. Lambda0/10 are frozen controls; inference is unchanged.

Lambda sweep | Test IC 73 | Cantera vs NN NRMSE

T0 = 1106.0 K P0 = 1.911 atm phi0 = 1.915 | Raw self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

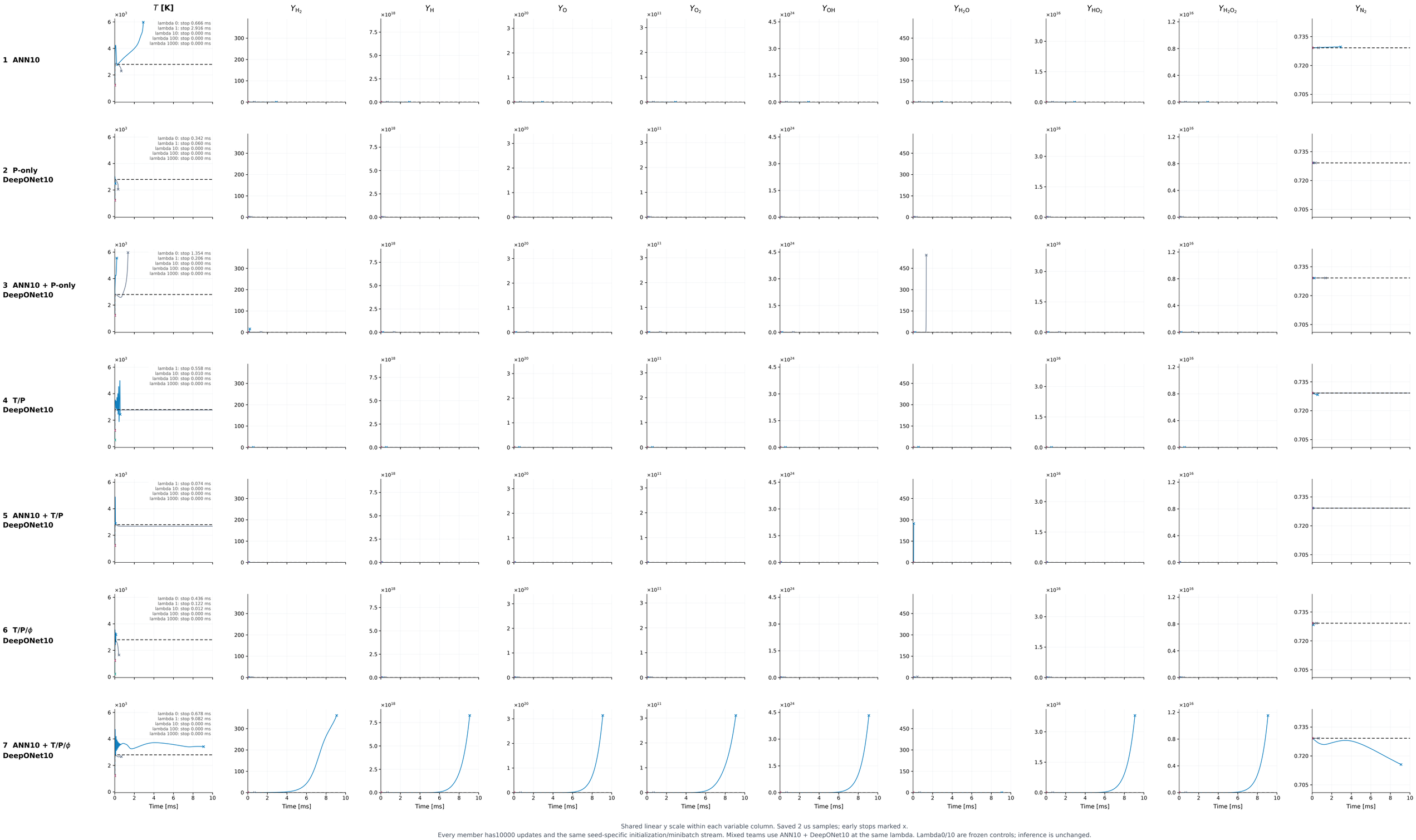
Model / same-state mean	lambda	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 0.918 ms
1 ANN10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.136 ms
1 ANN10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
1 ANN10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
1 ANN10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 3.434 ms
2 P-only DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.098 ms
2 P-only DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	**0**	0.06	0.5	0.694	2.01	0.0571	0.946	0.199	0.207	0.307	7.64e-16	Complete
3 ANN10 + P-only DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.210 ms
3 ANN10 + P-only DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	**0**	0.0213	0.271	0.332	0.788	0.495	0.316	0.107	1.23	1.2	7.64e-16	Complete
4 T/P DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.340 ms
4 T/P DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.660 ms
4 T/P DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	**0**	0.0125	0.0563	0.39	0.725	0.384	0.262	0.213	0.953	1.05	7.64e-16	Complete
5 ANN10 + T/P DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.178 ms
5 ANN10 + T/P DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 0.312 ms
6 T/P/ϕ DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.324 ms
6 T/P/ϕ DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.480 ms
6 T/P/ϕ DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	**0**	0.0314	0.167	0.638	1.09	0.399	0.587	0.172	0.993	1.07	7.64e-16	Complete
7 ANN10 + T/P/ϕ DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.150 ms
7 ANN10 + T/P/ϕ DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms

Exact user util.py defaults: norm="rms", reference=Cantera; equal time-sample weights, dimensionless ratio without x100.
N/A: undefined zero-reference RMS or empty common set. —: early stop, so no full-horizon NRMSE. Nonzero constant N2 is valid.
Every network has10000 updates. Lambda0/10 controls are reused; lambda1/100/1000 are fresh paired-seed models.
Every graph row includes all5 lambdas and the same Cantera reference.

Element soft-constraint weight sweep | Test IC 74

T_0 = 1222.9 K P_0 = 4.971 atm ϕ_0 = 1.766 | Raw self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference
lambda = 0 (10k)
lambda = 1 (10k)
lambda = 10 (10k)
lambda = 100 (10k)
lambda = 1000 (10k)



Shared linear y scale within each variable column. Saved 2 us samples; early stops marked x.
Every member has10000 updates and the same seed-specific initialization/minibatch stream. Mixed teams use ANN10 + DeepONet10 at the same lambda. Lambda0/10 are frozen controls; inference is unchanged.

Lambda sweep | Test IC 74 | Cantera vs NN NRMSE

T0 = 1222.9 K P0 = 4.971 atm phi0 = 1.766 | Raw self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	lambda	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 0.666 ms
1 ANN10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 2.916 ms
1 ANN10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
1 ANN10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
1 ANN10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 0.342 ms
2 P-only DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.060 ms
2 P-only DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 1.354 ms
3 ANN10 + P-only DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.206 ms
3 ANN10 + P-only DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	**0**	0.0146	0.0312	0.19	0.55	0.598	0.207	0.0463	1.03	0.965	1.83e-15	Complete
4 T/P DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.558 ms
4 T/P DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.010 ms
4 T/P DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	**0**	0.0414	0.092	0.578	0.866	0.539	0.599	0.208	1.07	0.987	1.37e-15	Complete
5 ANN10 + T/P DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.074 ms
5 ANN10 + T/P DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 0.436 ms
6 T/P/ϕ DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.122 ms
6 T/P/ϕ DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.012 ms
6 T/P/ϕ DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 0.678 ms
7 ANN10 + T/P/ϕ DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 9.082 ms
7 ANN10 + T/P/ϕ DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms

Exact user util.py defaults: norm="rms", reference=Cantera; equal time-sample weights, dimensionless ratio without x100.
N/A: undefined zero-reference RMS or empty common set. —: early stop, so no full-horizon NRMSE. Nonzero constant N2 is valid.
Every network has10000 updates. Lambda0/10 controls are reused; lambda1/100/1000 are fresh paired-seed models.
Every graph row includes all5 lambdas and the same Cantera reference.

Element soft-constraint weight sweep | Test IC 76

T_0 = 1523.6 K P_0 = 3.626 atm ϕ_0 = 1.349 | Raw self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference
lambda = 0 (10k)
lambda = 1 (10k)
lambda = 10 (10k)
lambda = 100 (10k)
lambda = 1000 (10k)

1 ANN10

2 P-only DeepONet10

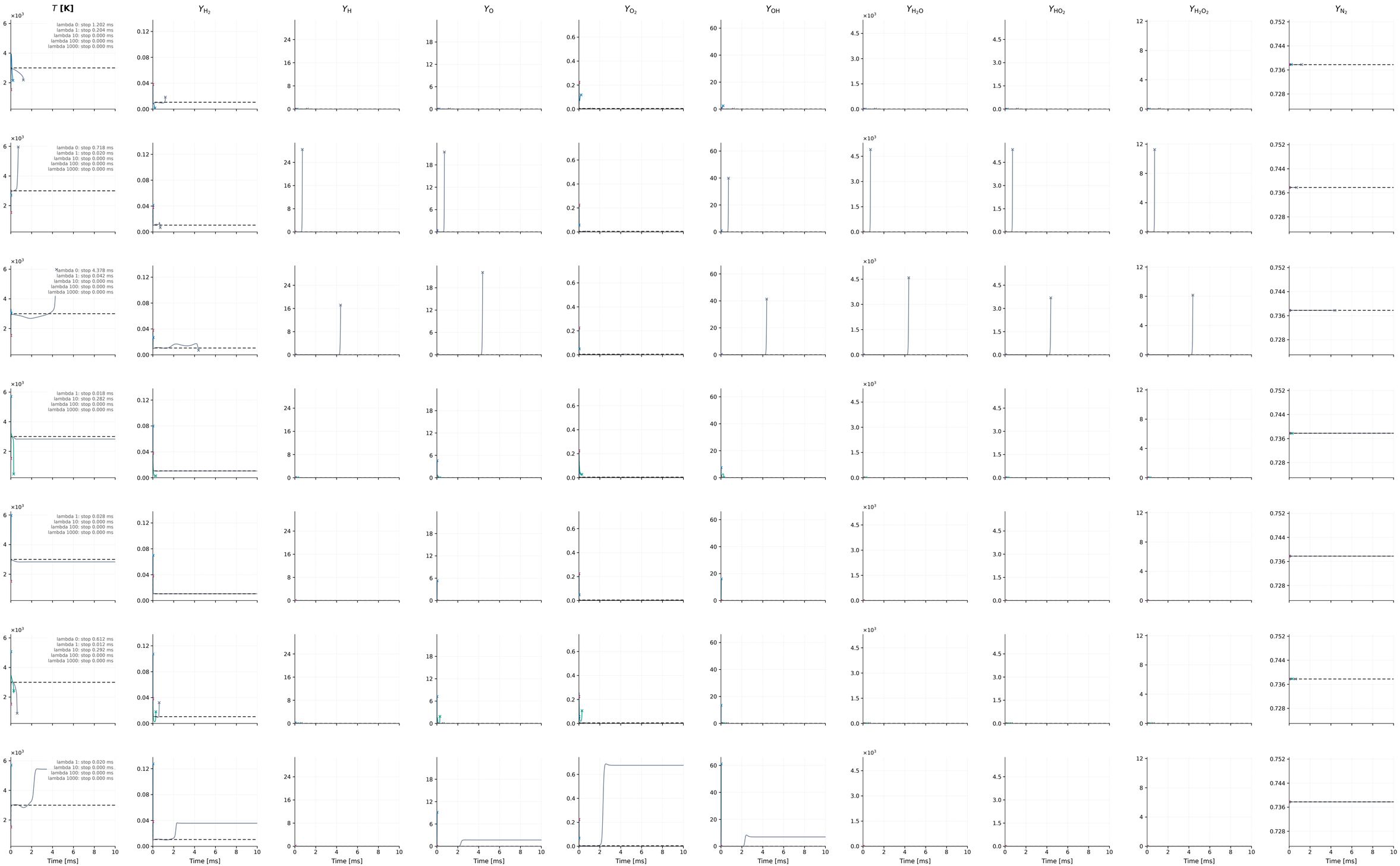
3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10



Shared linear y scale within each variable column. Saved 2 us samples; early stops marked x.
Every member has10000 updates and the same seed-specific initialization/minibatch stream. Mixed teams use ANN10 + DeepONet10 at the same lambda. Lambda0/10 are frozen controls; inference is unchanged.

Lambda sweep | Test IC 76 | Cantera vs NN NRMSE

T0 = 1523.6 K P0 = 3.626 atm phi0 = 1.349 | Raw self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

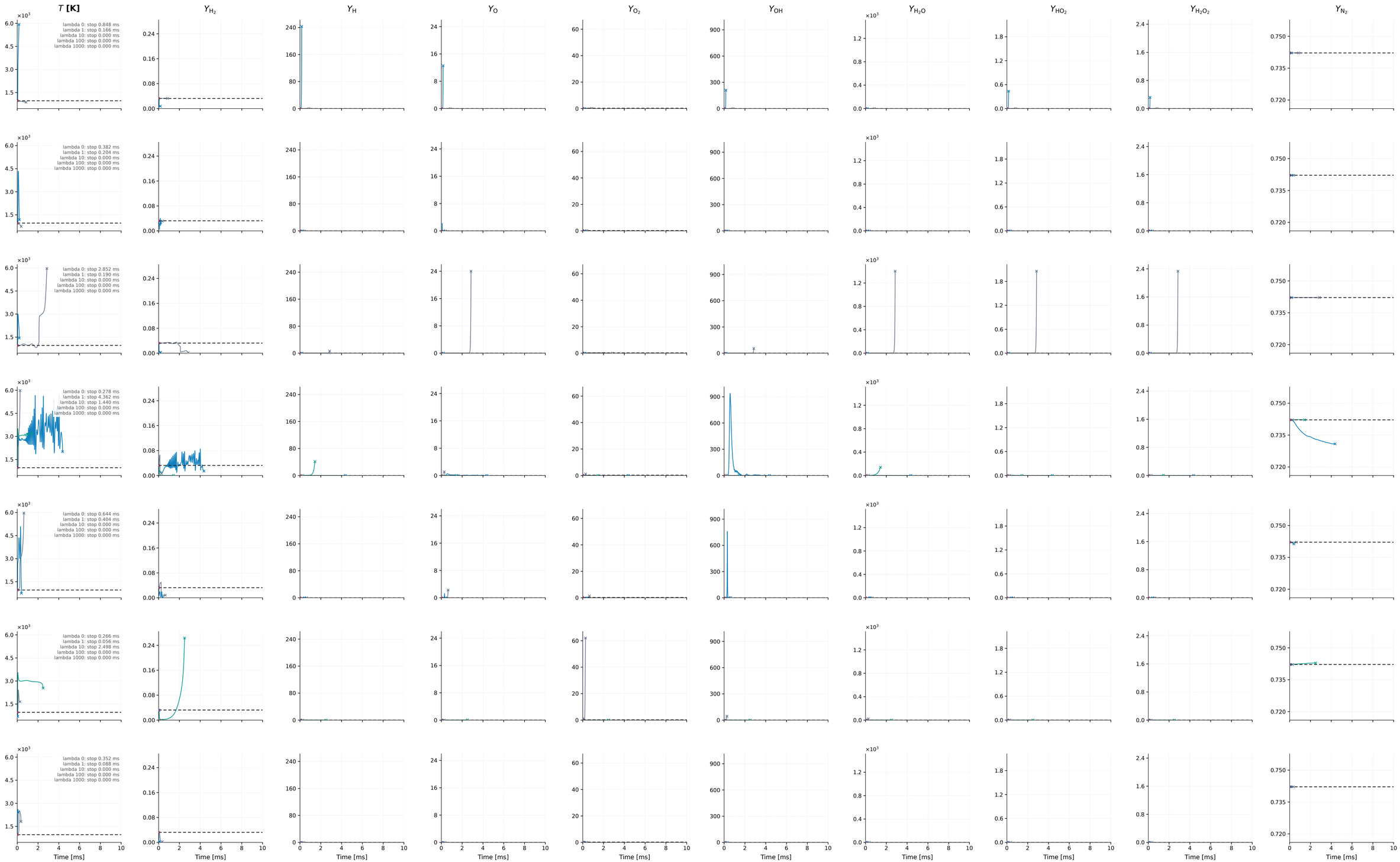
Model / same-state mean	lambda	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 1.202 ms
1 ANN10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.204 ms
1 ANN10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
1 ANN10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
1 ANN10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 0.718 ms
2 P-only DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.020 ms
2 P-only DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 4.378 ms
3 ANN10 + P-only DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.042 ms
3 ANN10 + P-only DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	**0**	0.0553	0.0153	0.391	0.634	0.466	0.448	0.0778	0.653	0.534	6.02e-16	Complete
4 T/P DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.018 ms
4 T/P DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.282 ms
4 T/P DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	**0**	0.0546	8.51e-03	0.396	0.647	0.485	0.475	0.0889	0.671	0.58	6.02e-16	Complete
5 ANN10 + T/P DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.028 ms
5 ANN10 + T/P DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 0.612 ms
6 T/P/ϕ DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.012 ms
6 T/P/ϕ DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.292 ms
6 T/P/ϕ DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	**0**	0.714	2.11	1.21	590	89.7	351	0.664	27.1	163	6.02e-16	Complete
7 ANN10 + T/P/ϕ DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.020 ms
7 ANN10 + T/P/ϕ DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms

Exact user util.py defaults: norm="rms", reference=Cantera; equal time-sample weights, dimensionless ratio without x100.
N/A: undefined zero-reference RMS or empty common set. —: early stop, so no full-horizon NRMSE. Nonzero constant N2 is valid.
Every network has10000 updates. Lambda0/10 controls are reused; lambda1/100/1000 are fresh paired-seed models.
Every graph row includes all5 lambdas and the same Cantera reference.

Element soft-constraint weight sweep | Test IC 78

T_0 = 962.3 K P_0 = 2.443 atm ϕ_0 = 1.140 | Raw self-fed rollout, $\Delta t = 2\,\mu$s





Shared linear y scale within each variable column. Saved 2 us samples; early stops marked x.
Every member has10000 updates and the same seed-specific initialization/minibatch stream. Mixed teams use ANN10 + DeepONet10 at the same lambda. Lambda0/10 are frozen controls; inference is unchanged.

Lambda sweep | Test IC 78 | Cantera vs NN NRMSE

T0 = 962.3 K P0 = 2.443 atm phi0 = 1.140 | Raw self-fed, 10 ms

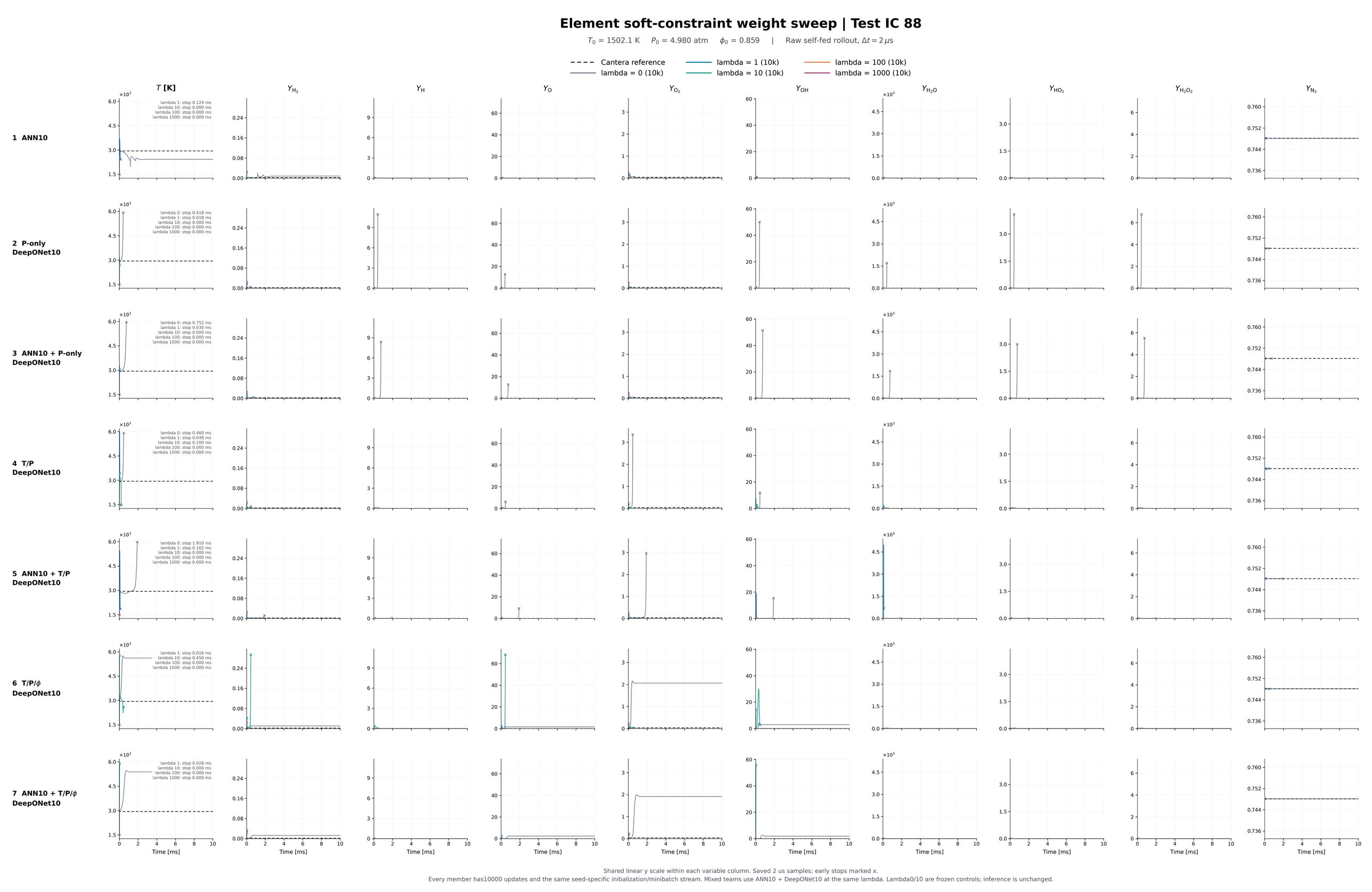
$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	lambda	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 0.848 ms
1 ANN10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.166 ms
1 ANN10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
1 ANN10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
1 ANN10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 0.382 ms
2 P-only DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.204 ms
2 P-only DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 2.852 ms
3 ANN10 + P-only DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.190 ms
3 ANN10 + P-only DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 0.278 ms
4 T/P DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 4.362 ms
4 T/P DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 1.440 ms
4 T/P DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 0.644 ms
5 ANN10 + T/P DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.404 ms
5 ANN10 + T/P DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 0.266 ms
6 T/P/ϕ DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.056 ms
6 T/P/ϕ DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 2.498 ms
6 T/P/ϕ DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 0.352 ms
7 ANN10 + T/P/ϕ DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.088 ms
7 ANN10 + T/P/ϕ DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms

Exact user util.py defaults: norm="rms", reference=Cantera; equal time-sample weights, dimensionless ratio without x100.
N/A: undefined zero-reference RMS or empty common set. —: early stop, so no full-horizon NRMSE. Nonzero constant N2 is valid.
Every network has10000 updates. Lambda0/10 controls are reused; lambda1/100/1000 are fresh paired-seed models.
Every graph row includes all5 lambdas and the same Cantera reference.

Element soft-constraint weight sweep | Test IC 88

T_0 = 1502.1 K P_0 = 4.980 atm ϕ_0 = 0.859 | Raw self-fed rollout, $\Delta t = 2\,\mu s$



Shared linear y scale within each variable column. Saved 2 us samples; early stops marked x.
Every member has10000 updates and the same seed-specific initialization/minibatch stream. Mixed teams use ANN10 + DeepONet10 at the same lambda. Lambda0/10 are frozen controls; inference is unchanged.

Lambda sweep | Test IC 88 | Cantera vs NN NRMSE

T0 = 1502.1 K P0 = 4.980 atm phi0 = 0.859 | Raw self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	lambda	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**0**	0.165	2.68	0.929	0.95	0.951	0.9	0.71	0.956	0.948	8.90e-16	Complete
1 ANN10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.124 ms
1 ANN10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
1 ANN10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
1 ANN10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 0.418 ms
2 P-only DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.018 ms
2 P-only DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
2 P-only DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 0.752 ms
3 ANN10 + P-only DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.030 ms
3 ANN10 + P-only DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
3 ANN10 + P-only DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 0.460 ms
4 T/P DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.048 ms
4 T/P DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.200 ms
4 T/P DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
4 T/P DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	**0**	—	—	—	—	—	—	—	—	—	—	Stop 1.910 ms
5 ANN10 + T/P DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.102 ms
5 ANN10 + T/P DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
5 ANN10 + T/P DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	**0**	0.896	3.88	2.42	341	53.6	140	0.713	5.19	42.7	8.90e-16	Complete
6 T/P/ϕ DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.016 ms
6 T/P/ϕ DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.450 ms
6 T/P/ϕ DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
6 T/P/ϕ DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	**0**	0.807	5.08	5.33	500	48.7	88	0.136	6.07	29.2	8.90e-16	Complete
7 ANN10 + T/P/ϕ DeepONet10	**1**	—	—	—	—	—	—	—	—	—	—	Stop 0.028 ms
7 ANN10 + T/P/ϕ DeepONet10	**10**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	**100**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms
7 ANN10 + T/P/ϕ DeepONet10	**1000**	—	—	—	—	—	—	—	—	—	—	Stop 0.000 ms

Exact user util.py defaults: norm="rms", reference=Cantera; equal time-sample weights, dimensionless ratio without x100.
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